

Biomaterials

生物材料

南方科技大学材料科学与工程系2024年春季课程

Host Reactions to Biomaterials and Their Evaluation

- A: Implantation and Inflammation, and the foreign-body response/reaction
- B: The Immune response to foreign materials
- C: Wound healing and the presence of biomaterials
- D: Blood coagulation and blood-materials interactions
- E: device-related infections

Foreign Body Reaction/Response

- The presence of the implant changes the healing response, and this is called the Foreign Body Reaction (FBR).
- FBR consists of:
 - protein adsorption
 - macrophages
 - multinucleated **foreign body giant cells**, which are cells of monocyte or macrophage lineage fused together
 - fibroblasts
 - angiogenesis

Fibrosis and Fibrous Encapsulation

- End stage of healing response
- Usually four or more weeks after implantation
- A relatively acellular fibrous capsule
 - spindle shaped fibroblasts
 - small number of macrophages
- Presence of neutrophils suggests persisting inflammatory challenge
- Presence of **foreign body giant cell** suggests production of small particles by corrosion, depolymerization, or dissolution

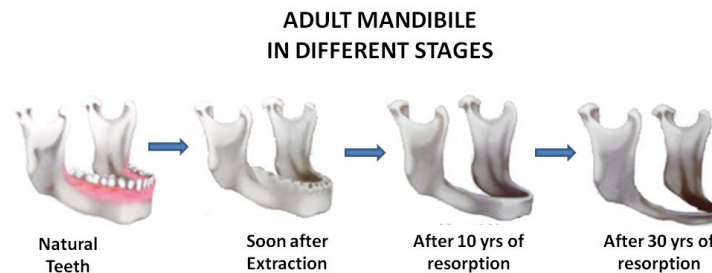
Foreign Body Giant Cells (FBGCs): multinucleated cells formed by fusion of monocytes/macrophages in an attempt to phagocytose biomaterials much larger than a single cell.

Foreign Body Response

– continuing presence of an implant may result in the attainment of a final steady-state condition called **resolution**

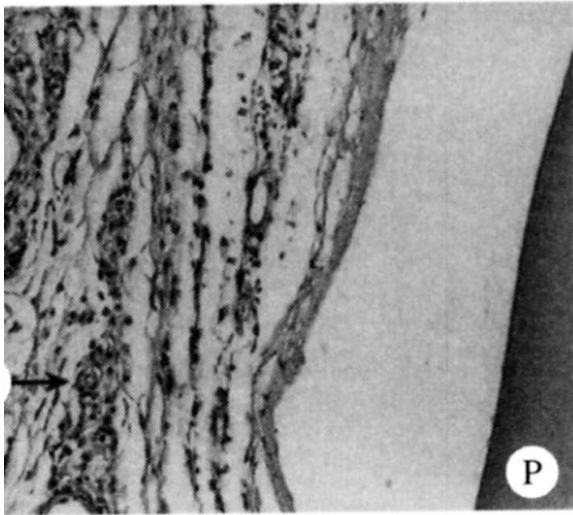
there are 4 possible outcomes for the implant :

- 1: Extrusion: material forced out of the body (e.g. splinter)
- 2: **Resorption**: material biodegrades, no fibrous capsule forms
- 3: Integration: implant and host tissue grow together (e.g. porous titanium implant in bone)
- 4: Encapsulation: implant surrounded by **fibrous tissue**

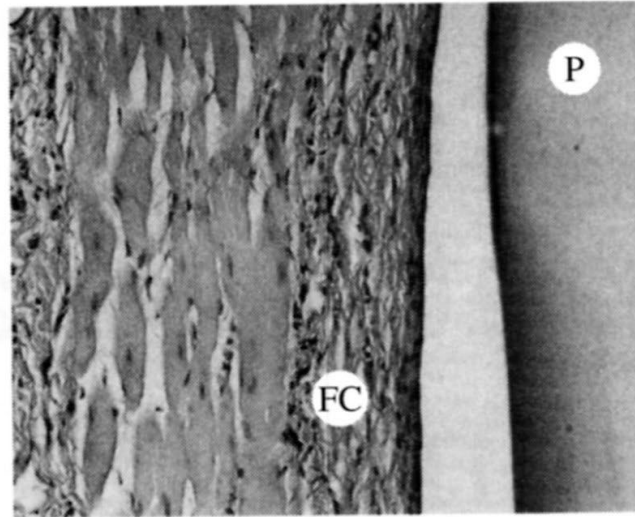


* Multiple missing teeth cause atrophy (wasting) of jaw bone
* Severe bone loss makes replacement of teeth difficult in later stages

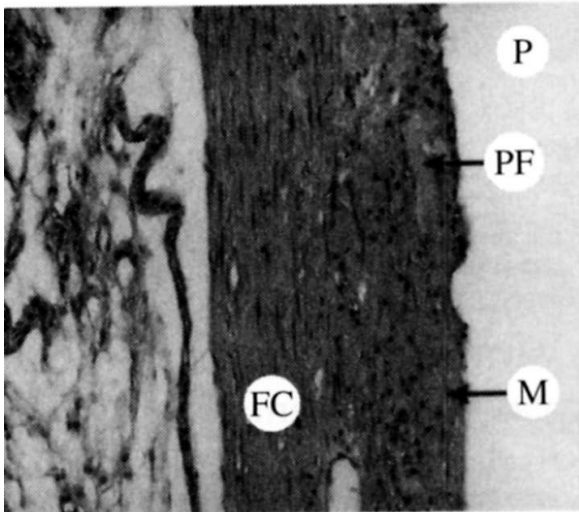
Fibrous Encapsulation



(a)



(b)



(c)

In vivo response to a **biodegradable**, polymeric biomaterial implanted in a rat for 12 weeks. (a) 4 days (b) 3 weeks (c) 12 weeks. P indicates polymer, or space left by polymer; N: neutrophils, FC: fibrous capsule, M: macrophages, PF: polymer fragments embedded in fibrous capsule. Infiltration of neutrophils into implantation area is seen within a few days, followed by slower development of fibrous capsule surrounding implant. Because material is biodegradable, polymer fragmentation is present at later times.

Fibrous Encapsulation, cont.

Final stage of healing for implants made of **nondegradable** materials.

Steps in granulation tissue maturation:

- Presence of larger blood vessels
- Alignment of collagen fibers in response to local mechanical forces
- Collapse of capsule surrounding implant and formation of a scar

Factors affecting capsule formation:

- Degree of original injury during implantation
- Amount of subsequent cell death
- Location of implant site
- Degradation time of implant

Factors affecting capsule thickness:

- Amount and composition of small particulates produced
- Mechanical factors at implant site
- Shape of implant
- Electrical currents

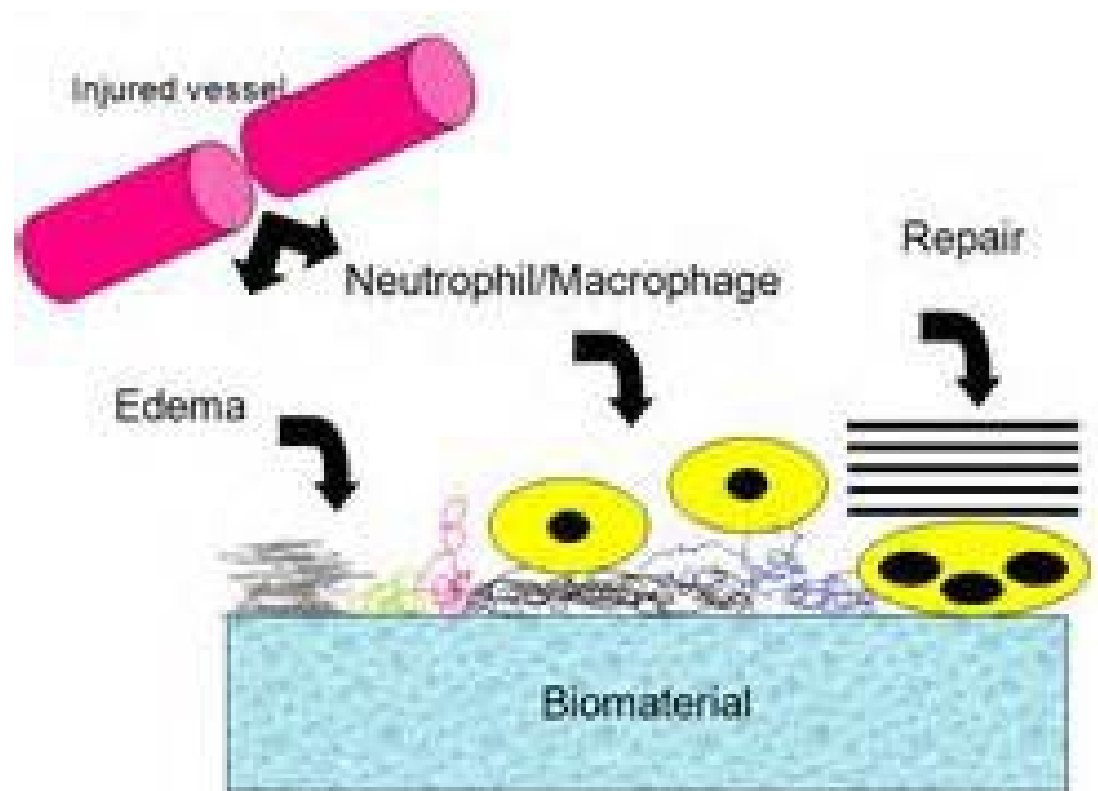
Foreign Body Response (affecting factors)

Factors affecting Foreign
Body Composition:

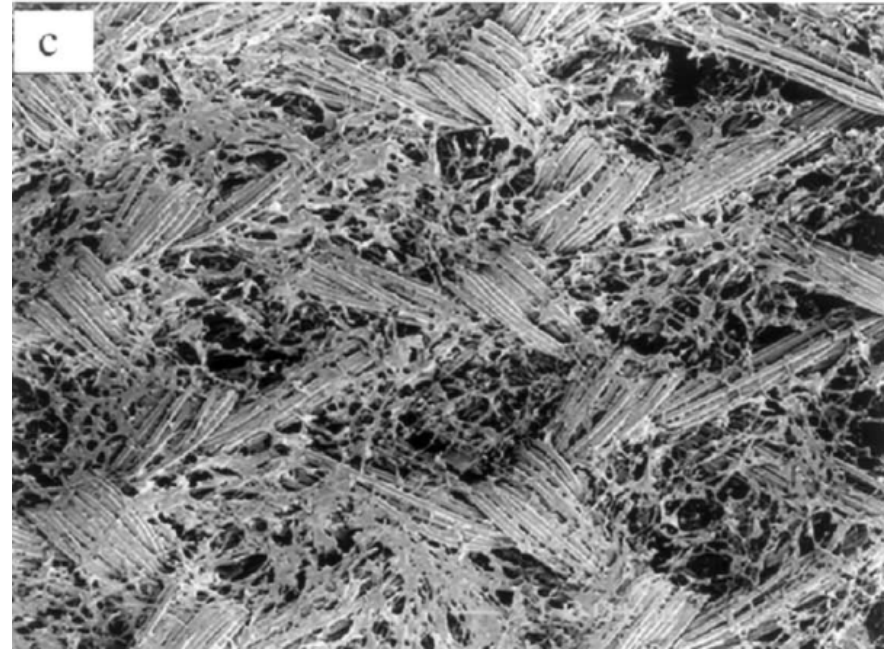
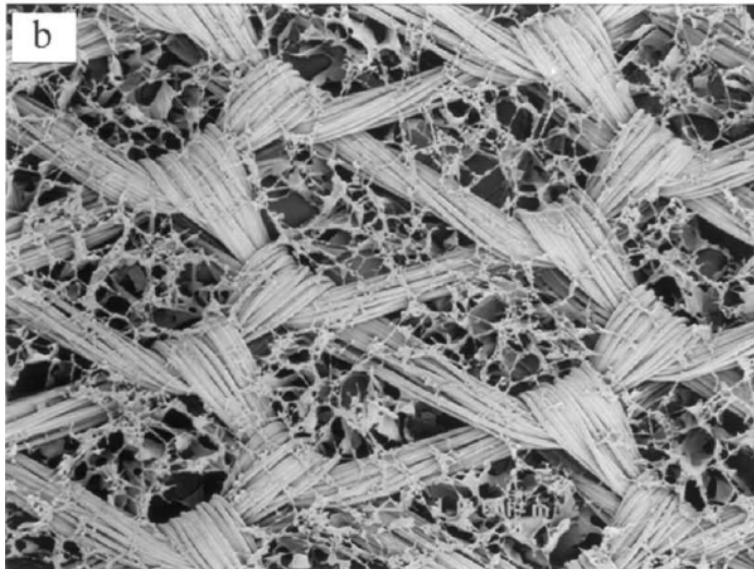
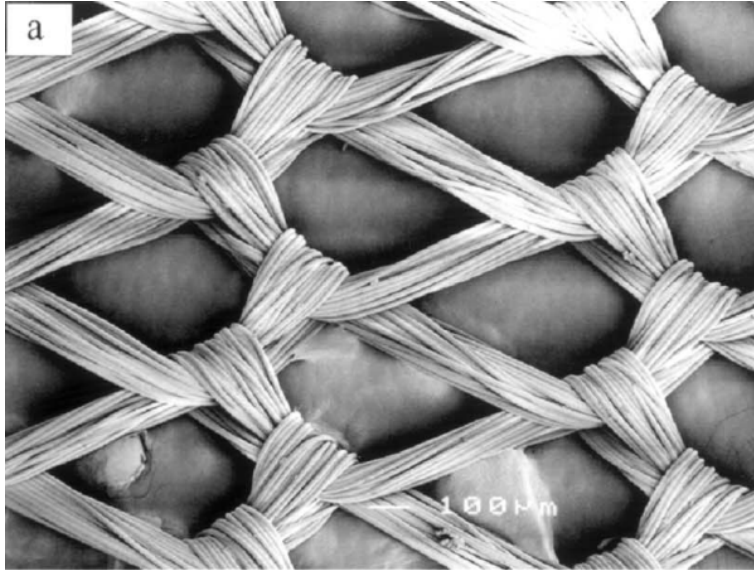
1. Topography
2. Surface Chemistry

Factors affecting Foreign
Body Reaction:

1. Shape
2. Surface/Volume Ratio



Surface-volume ratio helps cell culture

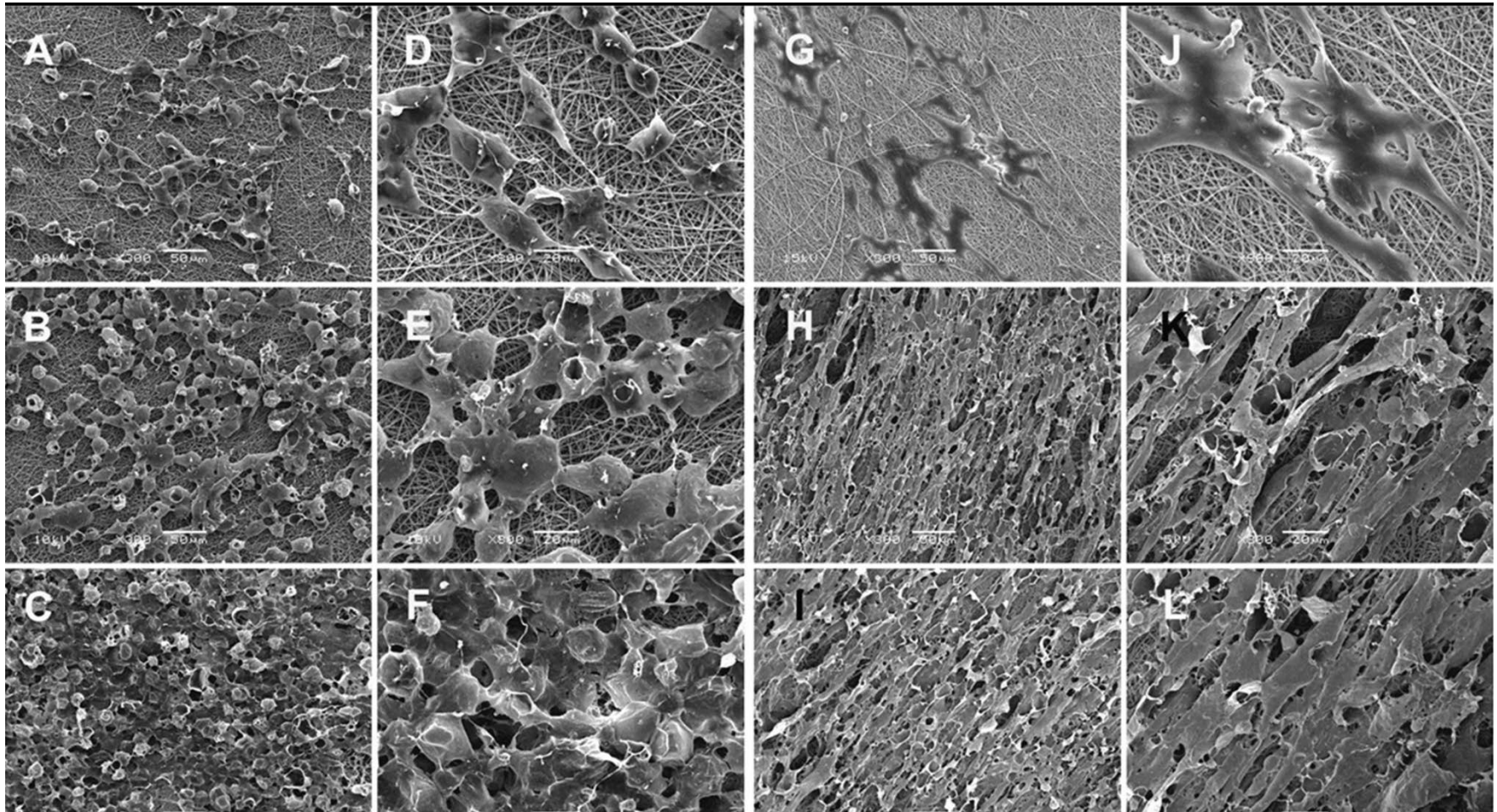


SEM photomicrographs of PLGA knitted mesh (a), PLGA -collagen hybrid mesh (b), and human fibroblasts cultured in the hybrid mesh for one day (c). [Macromol. Biosci. 2002, 2, 67 – 77.](#)

Surface-volume ratio helps cell culture

Human vascular endothelial cells (ECs)

Human coronary artery smooth muscle cells (SMC)



SEM photomicrographs showing adherence and proliferation of ECs and SMCs cultured on S-silk scaffolds for 1, 7, and 14 days. (A–C, G–I) **scale bars = 50 μm** ; (D–F, J–L) **scale bars = 20 μm** . *Journal of Nanomaterials*, Volume 2013, Article ID 495708.

Wound Healing and the Presence of Biomaterials

Topics:

- Formation of Granulation Tissue
- Foreign Body Reaction
- Fibrous Encapsulation
- Chronic Inflammation
- Types of Implant Resolution
- Repair vs. Regeneration

Wound Healing

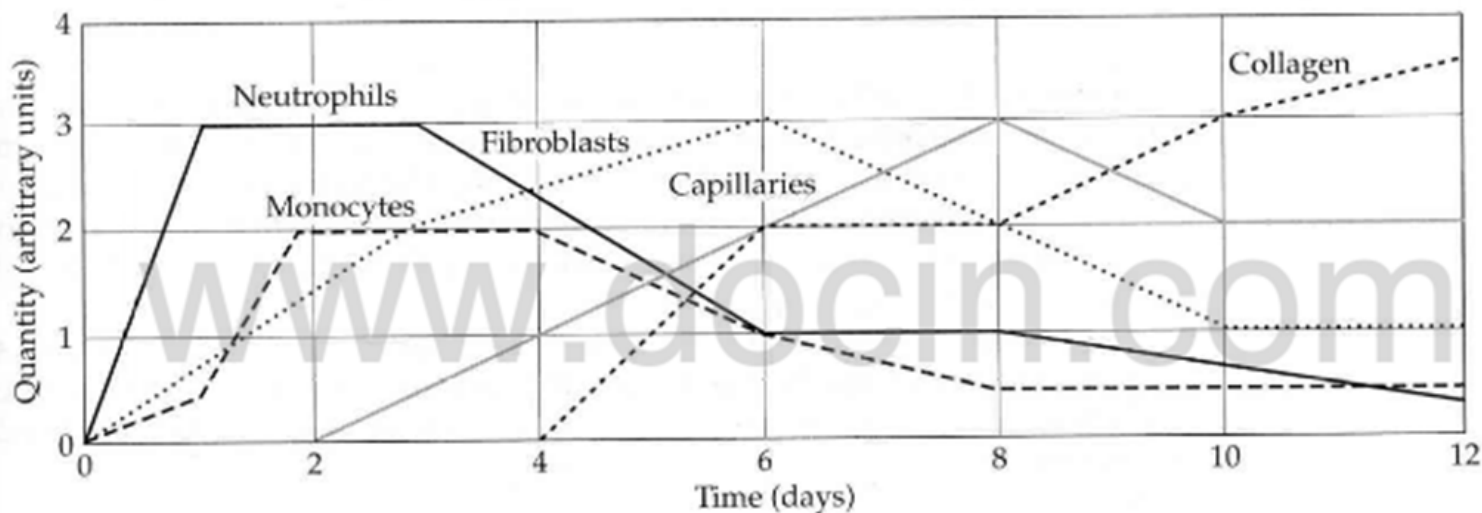
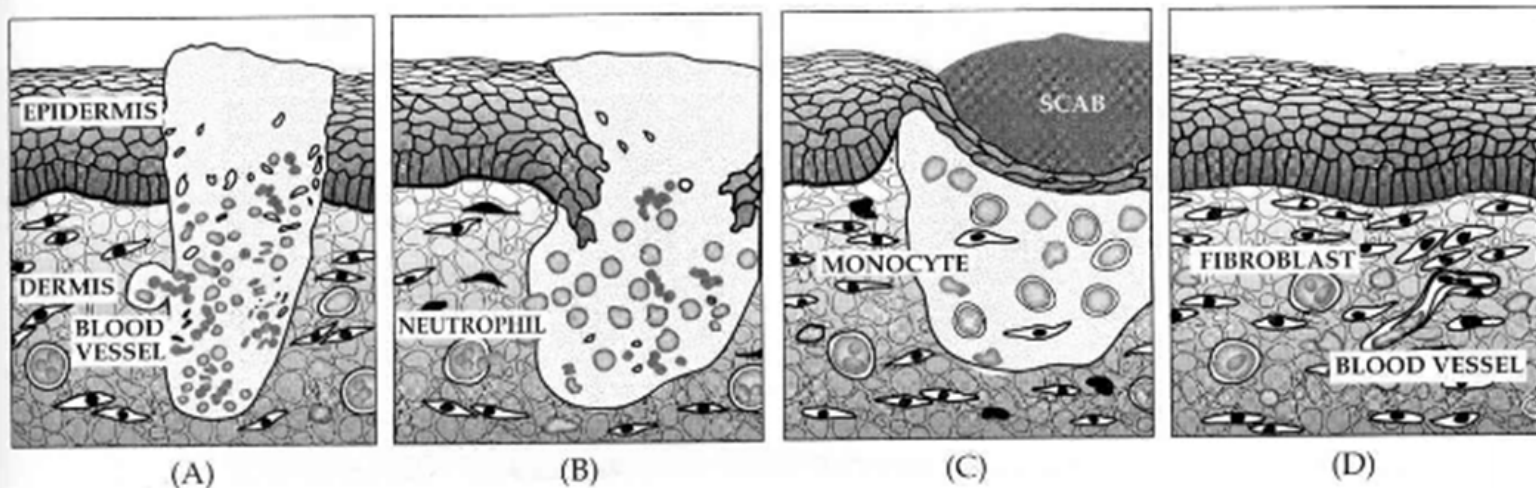


Figure 3.3 Wound healing in adult skin. (A) hemostasis of the wound occurs; (B) thrombosis (clotting) forms a fibrin plug; (C) inflammation (migration of white blood cells); (D) formation of granulation tissue with new blood vessels.

Responses following injury:

1. Blood clotting and formation of fibrin network

2. Acute inflammation:

- activation of neutrophils
- phagocytosis of foreign bodies
- release of hyaluronic acid and glycosaminoglycan (chemoattractants) into ECM.

3. Inflammatory response:

influx of fibroblasts into ECM

beginnings of granulation tissue formation

generation of new blood vessels

deposition of type III collagen fibers (thin and randomly oriented)

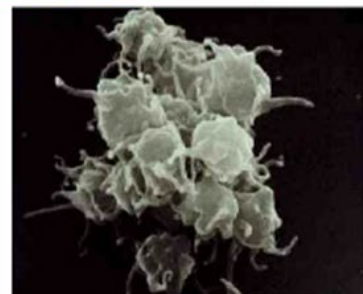
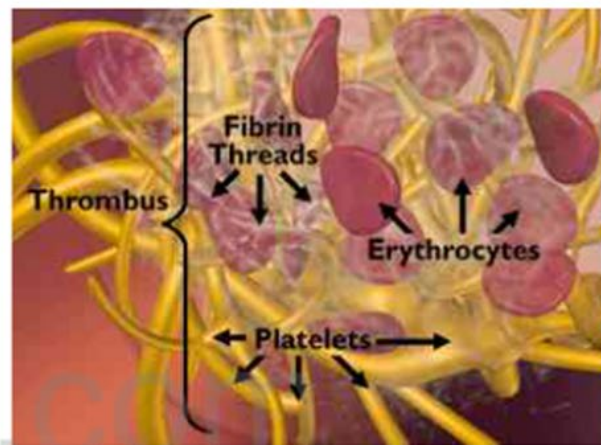
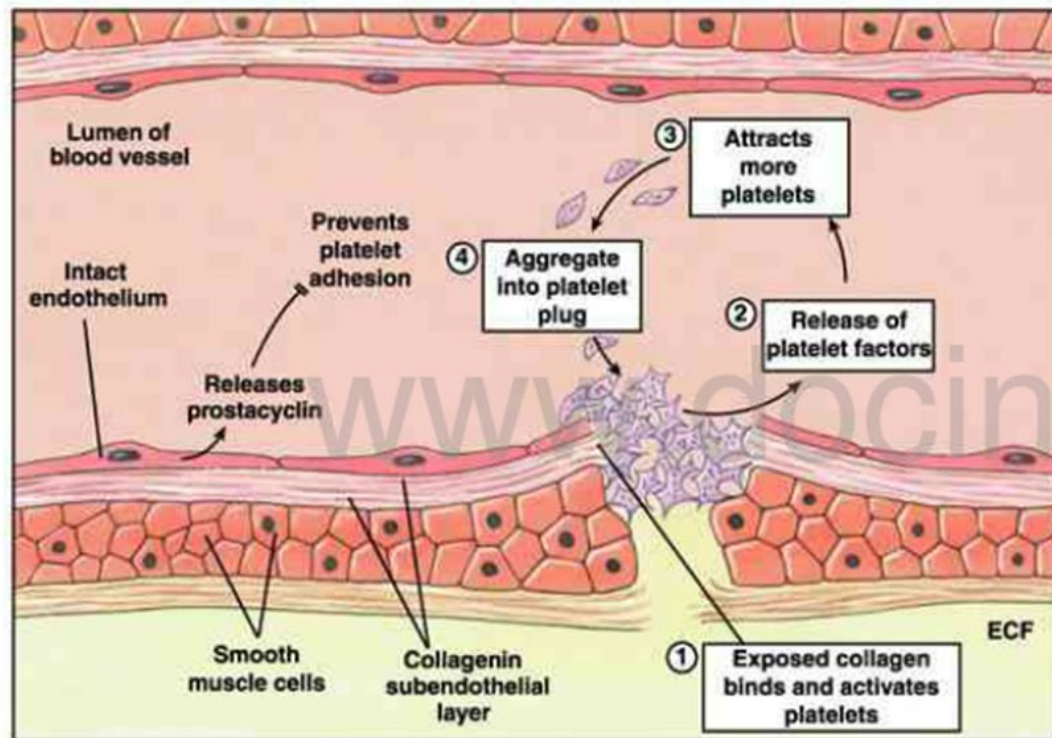
fibrin clot is dissolved, enzymes released and phagocytosis continues

4. Remodeling and scar formation:

- Type III collagen replaced by type I collagen: collagen bundles are larger and oriented with principal lines of stress in tissue
- Increased amounts of chemicals such as chondroitin and dermatan sulfate
- Scar tissue continues to form for several months
- Blood vessels that are unattached are resorbed
- Scar becomes pale and avascular

Clotting or Thrombosis

- blood coagulation or clotting is the blood response to damaged blood vessels
- objective is to form a patch that isolates the leak and stops blood loss



Granulation Tissue: characterized by a pebbly, granular appearance caused by the creation of many *vascular buds* sprouting from existing blood vessels. This process is called ***neovascularization*** or ***angiogenesis***.

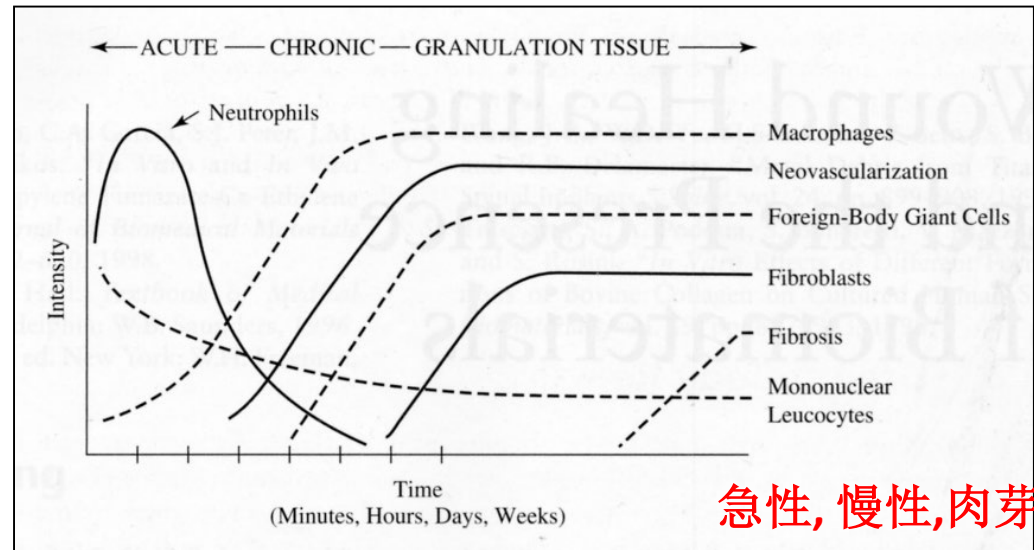


Fibroblasts: committed cell type found in many tissues. *Fibroblasts* synthesize and maintain connective tissues by producing and extracellular matrix (ECM) rich in collagen and proteoglycans. Fibroblasts with features of smooth muscle cells are called ***myofibroblasts*** and are responsible for wound contraction.

成纤维细胞, 肌成纤维细胞

Formation of Granulation Tissue

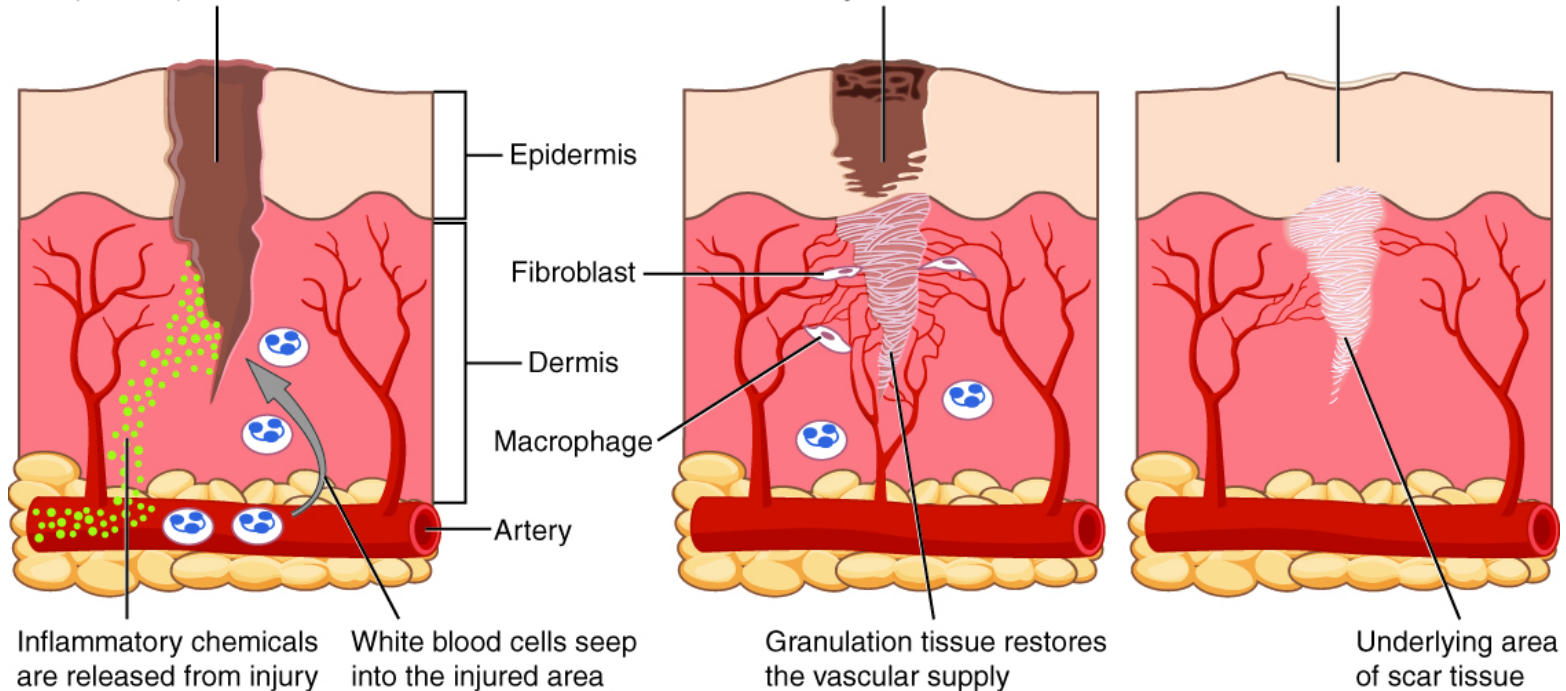
Wound-healing
response of the body
after injury or
biomaterial
implantation



Clotting occurs, caused by clotting proteins and plasma proteins, and a scab is formed

Epithelial cells multiply and fill in over the granulation tissue

Restored epithelium thickens; the area matures and contracts

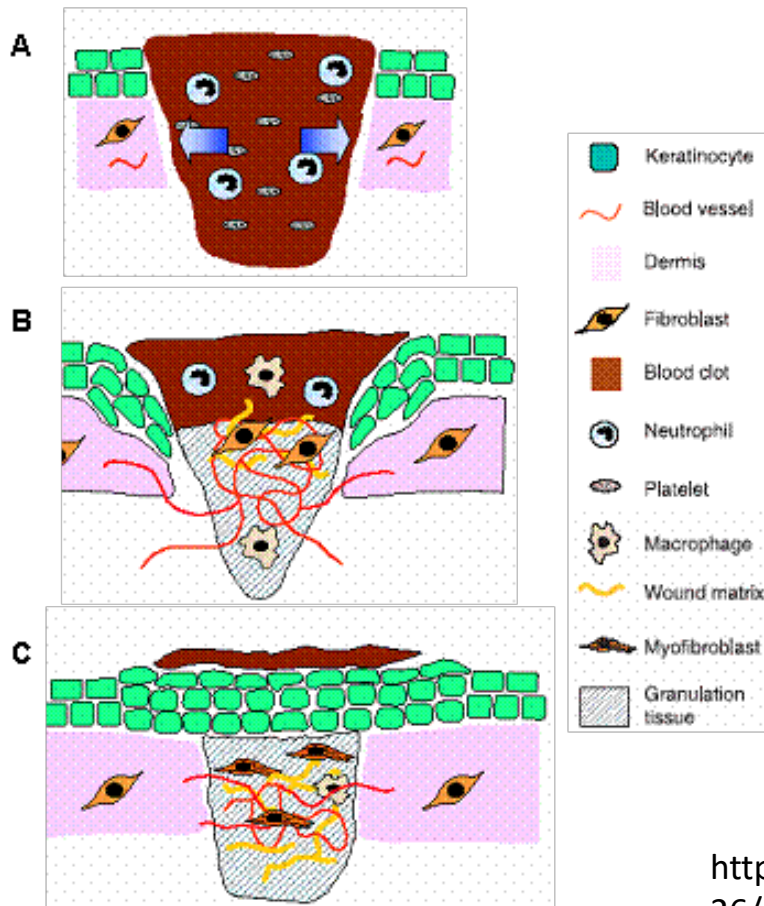


Repair vs. Regeneration

Wound healing in Skin

Repair involves healing of the internal *dermal* layer

Regeneration is regrowth of thin outer *epidermal* layer



Scheme of different stages of wound repair in mammals A: 12–24 h after injury the wounded area is filled with a blood clot. B: at days 3–7 after injury, endothelial cells migrate into the clot; they proliferate and form new blood vessels. Fibroblasts migrate into the wound tissue, where they proliferate and form extracellular matrix. The new tissue is called granulation tissue. Keratinocytes proliferate at the wound edge and migrate down the injured dermis and above the provisional matrix. C: 1–2 wk after injury the wound is completely filled with granulation tissue. Fibroblasts have transformed into myofibroblasts, leading to wound contraction and collagen deposition. The wound is completely covered with a neoepidermis. Modified from Werner and Grose, 2003.

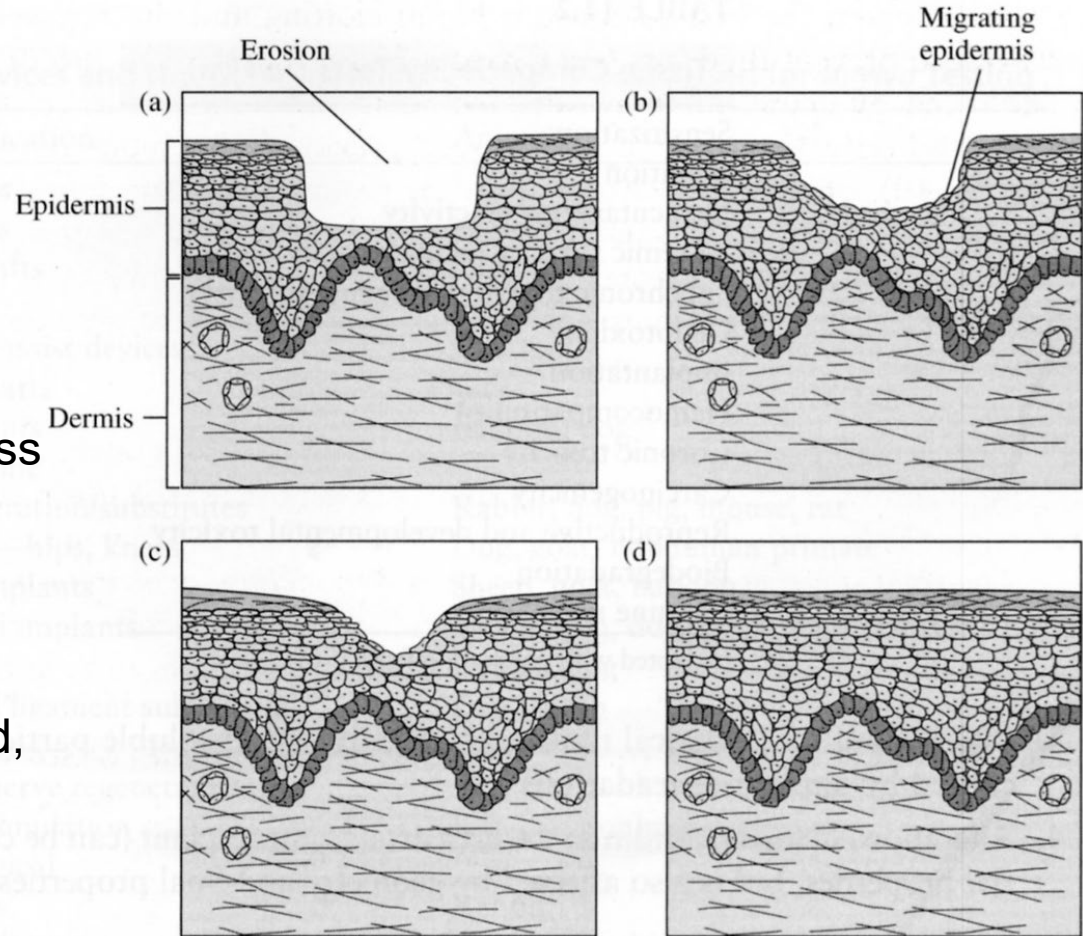
Skin Regeneration

In the epidermis, this process is called *reepithelization*.

Cells at edge of wound flatten to cover more of the wound, releasing attachment to ECM to migrate across wound

Epithelial cells gradually cover the entire wound site

ECM attachments are reestablished, and cells recover original shape



Blood Coagulation and Blood-Material Interactions

Blood is the first “tissue” that any surface of most implants will contact

Implanting a device causes injury which results in bleeding thus starting the wound healing cascade

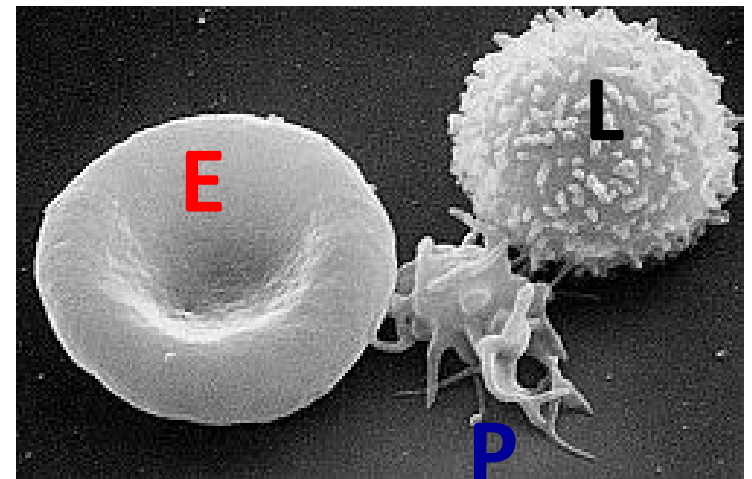
Injury -- **Coagulation** -- **Inflammation** -- Repair and Remodeling

Blood is a mixture of:

Plasma: The non-cell containing component of blood – water, salts and proteins

Numerous cell types: **Platelets (P)**, Leukocytes (L), **Erythrocytes (red blood cells; E)**

Blood cells originate from the bone marrow by a process known as **hematopoiesis**



Red blood cells

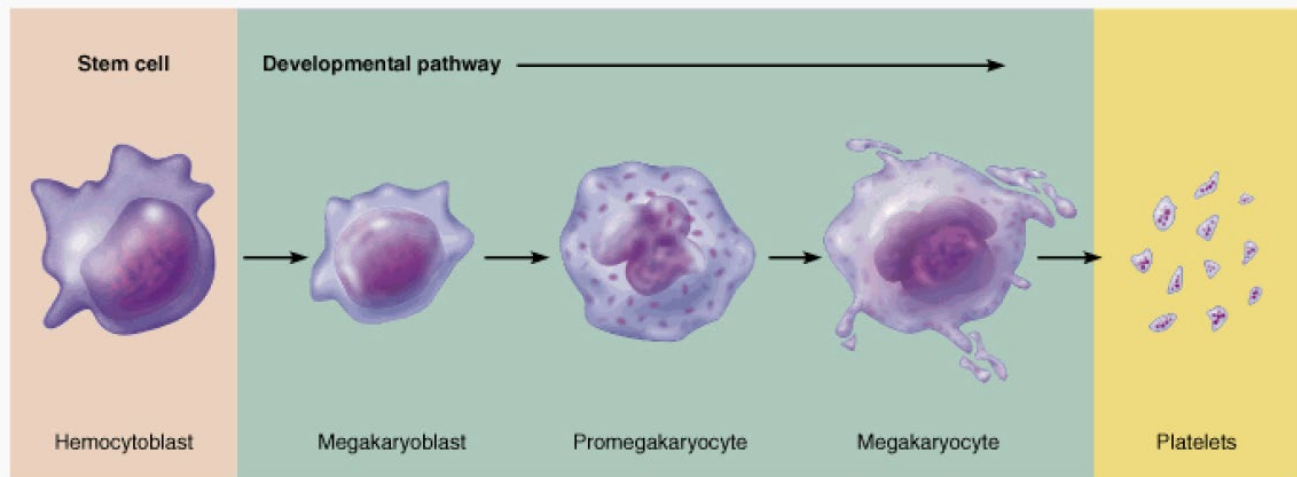
- Have no nucleus, and so are extremely deformable (due to the flow properties we just saw)
 - Membrane effects in a red blood cell:
<http://www.youtube.com/watch?v=Ym1rvwP-po4>
- The ability to deform is crucial to the function of RBCs (**Why?**)



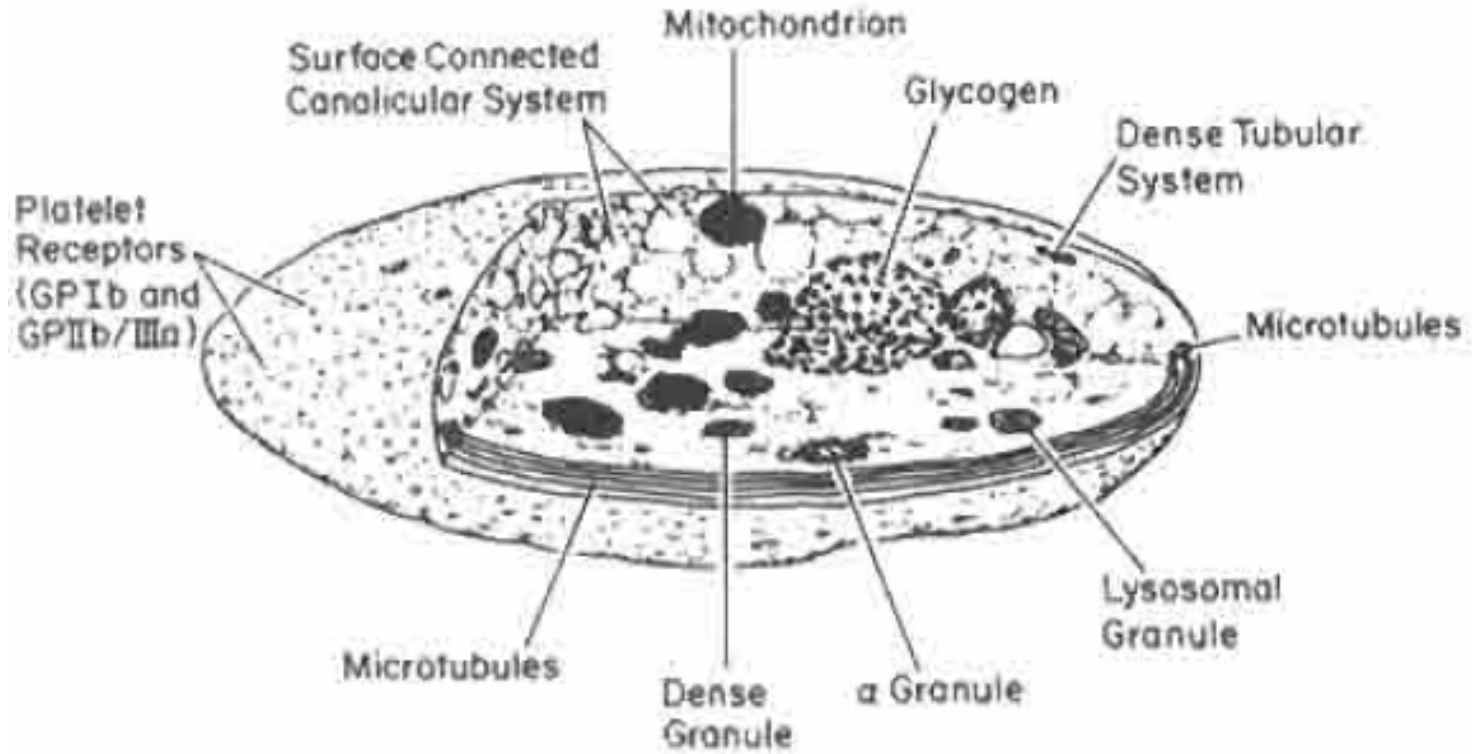
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Platelets ("little plates") are non-nucleated, disk-shaped cells having a diameter of 3 – 4 μm , and an average volume of $10 \times 10^{-9} \text{ mm}^3$.

Platelets are produced in the bone marrow, circulate at an average concentration of about 250,000 cells/ μL and occupy approximately 0.3 % of the total blood volume. In contrast, red cells typically circulate at 5×10^6 cells per microliter, and may comprise 40—50% of the total blood volume. As described later, platelet functions are designed to: (1) initially arrest bleeding through formation of platelet plugs, and (2) stabilize platelet plugs by catalyzing coagulation reactions, leading to the formation of fibrin.

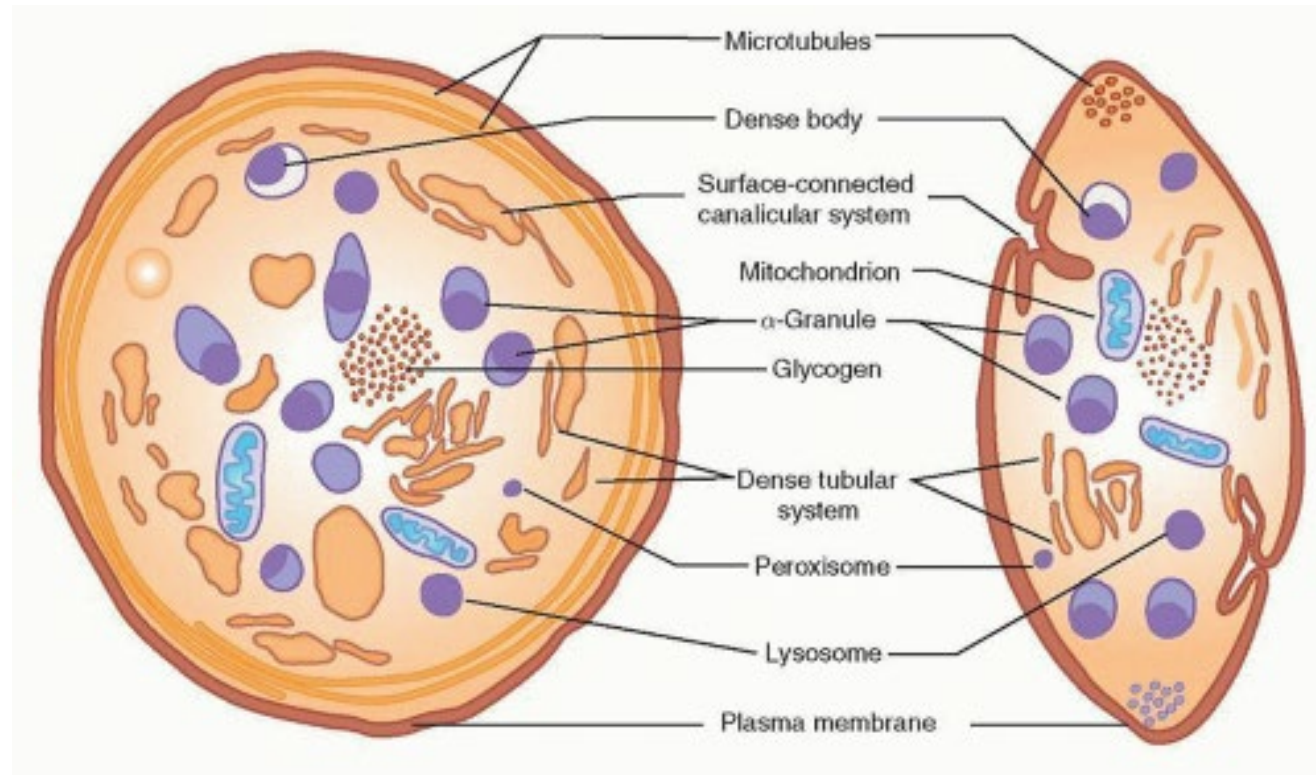


Platelets



In the normal (non-stimulated) state, the platelet discoid shape is maintained by a circumferential bundle (cytoskeleton) of microtubules. The external surface coat of the platelet contains membrane-bound receptors (**e.g., glycoproteins Ib and IIb/IIIa**) that mediate the **contact reactions of adhesion (platelet-surface)** and aggregation (platelet-platelet). The membrane also provides **a phospholipid surface which accelerates coagulation reactions**, and form a spongy, canal-like open network which represents an expanded reactive surface to which plasma factors are selectively adsorbed.

Platelets

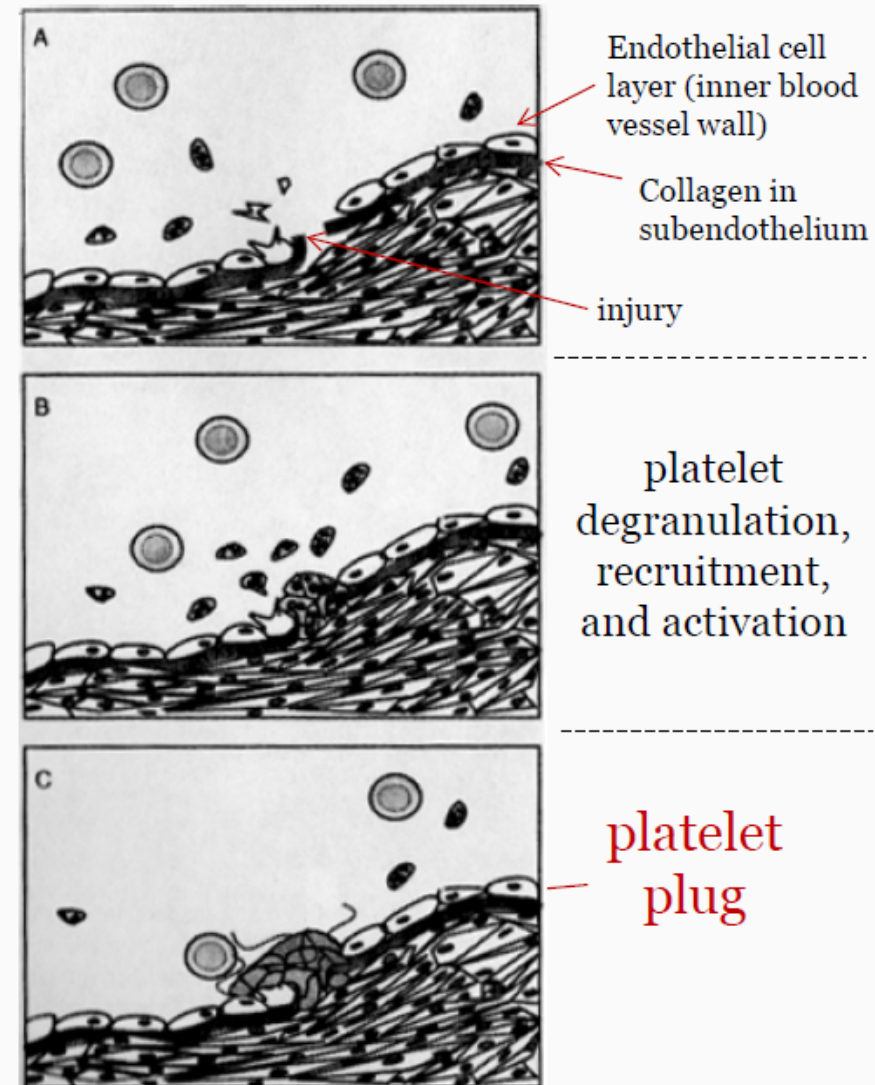


In addition to membranous components (plasma membrane, surface-connected canalicular system, and dense tubular system), mitochondria, microtubules, and glycogen, four types of storage organelles are identified: α -granules, dense bodies, lysosomes, and microperoxisomes. Whereas the first two can be identified morphologically, microperoxisomes and lysosomes are recognizable only by cytochemical stains. (From Bentfeld-Barker ME, Bainton DF. Identification of primary lysosomes in human megakaryocytes and platelets. *Blood* 1982;59:472-481)

Aggregations and Coagulation

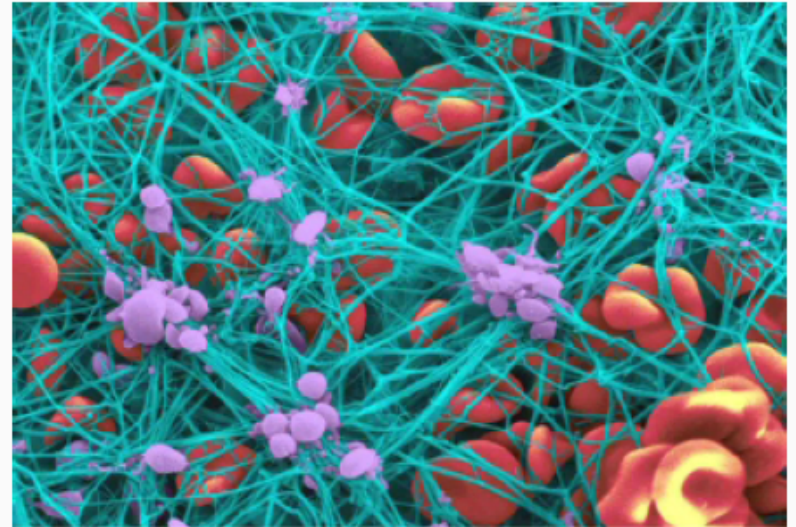
- Upon injury, platelets adhere to a surface → contractile proteins (**actin and myosin**) tighten → platelet flattens and forms **pseudopodia** (“legs”)
 - Typically platelets adhere to the connective tissue exposed when the endothelium of a blood vessel is ruptured (right)
 - However, they also adhere to the surface of many man-made biomaterials
- Contraction causes platelets to **degranulate**, releasing **ADP** and **thromboxane A_2**
 - These are potent activators of platelets, recruiting more locally
 - **glycoprotein IIb/IIIa** is also upregulated for each platelet

“two-pronged approach”: more platelets, plus a higher affinity per platelet



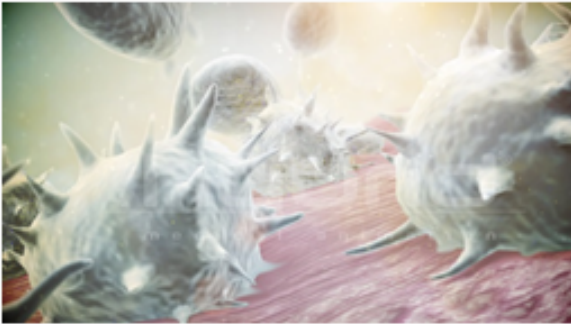
Aggregations and Coagulation

- The mass of aggregated platelets further contract actin/myosin, drawing the edges of the injury together
- An end product of coagulation is production of sticky threads of the protein **fibrin**
 - Fibrin threads attach to and help consolidate the platelet plug
 - Also trap nearby erythrocytes
- At this point the platelet plug is a **blood clot**



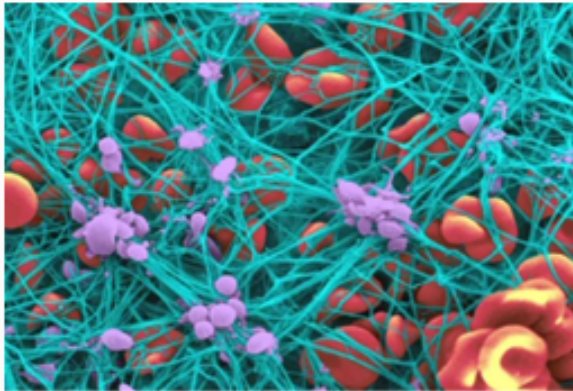
Colorized scanning electron microscope (SEM) of a whole blood clot. Fibrin fibers are blue, platelet aggregates purple, and red blood cells red (Source: Yuri Veklich and John W. Weisel, University of Pennsylvania School of Medicine)

Aggregations vs. Coagulation



<http://www.hybridmedicalanimation.com/>

Platelet aggregation – the accumulation of platelets to the site of a wound to form a platelet plug



Coagulation – the activation of clotting factors following platelet aggregation, ultimately resulting in a thrombus or a blood clot

Biomaterials can be affected both by aggregation and coagulation

Anticoagulants and fibrinolysis (纤维蛋白溶解)

- **Nonspecific** biologic control mechanism

Blood flow: passing blood volume dilutes coagulants and removes them from the local injury area

- **Specific** anticoagulants

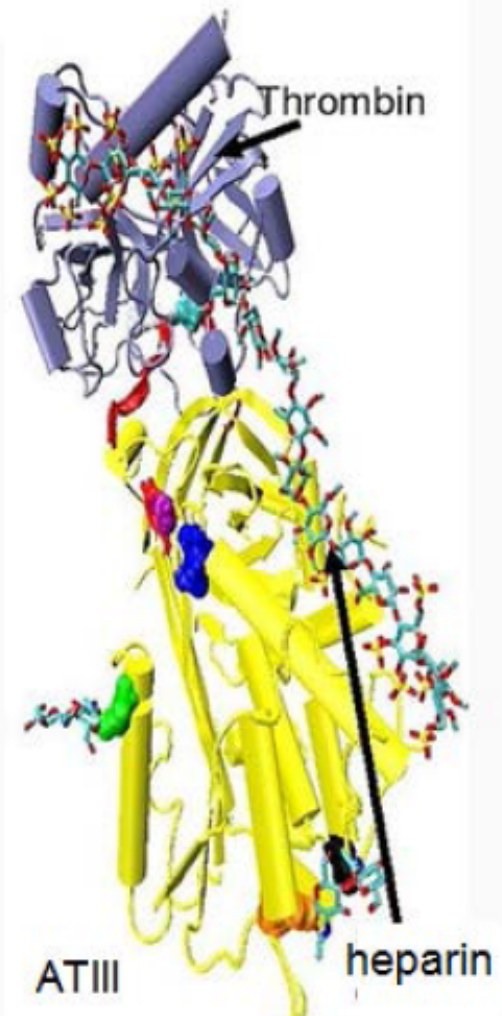
- **endothelium**: smooth and “slimy” nature of glycocalyx mucopolysaccharide layer on the blood contacting surface

血栓调节蛋白 – **thrombomodulin**: binds thrombin, making it unavailable for further coagulation cascades

纤维蛋白 – **fibrin**: ~85 – 90% of thrombin formed during coagulation absorbs to the newly polymerized fibrin threads, localizing the coagulation

- **antithrombin III (ATIII)**: serum protein that binds and inactivates free thrombin

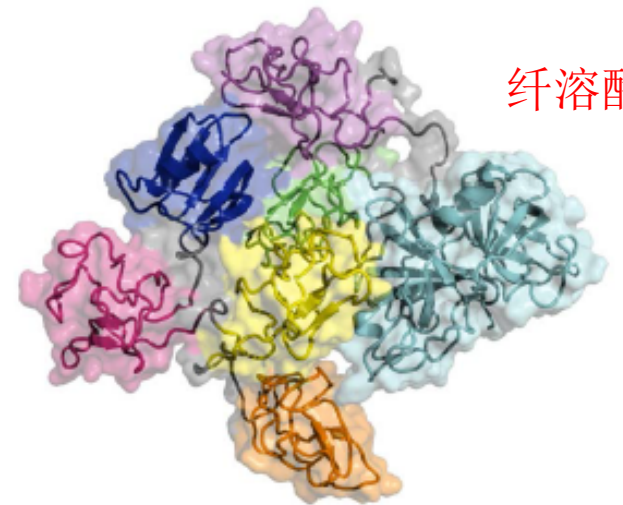
Thrombin: 凝血酶



抗凝

Removal of a blood clot via fibrinolysis

- Clots are actively removed by enzymatic digestion that includes a time delay of 24 – 48 hours
 - During coagulation, **plasminogen**, which is circulating in the plasma, is trapped in the forming clot
 - Injured tissue releases **tissue plasminogen activator** that converts plasminogen to **plasmin** over 1 – 2 days
- 纤溶酶 – **Plasmin** is a proteolytic enzyme that digests the fibrin threads of the clot and other pro-coagulants



纤溶酶原

Biomaterials, devices, and thrombosis

- Normal, healthy endothelium does not induce coagulation!
- Such is the goal of implantable blood-contacting biomaterials
 - Think of deleterious effects of an implant as in two categories:

Harmful to the implant or its function

- Adsorption of blood components (e.g., proteins) onto the biomaterial
- Adsorption of blood cells onto the biomaterial
- Tissue growth around the biomaterial

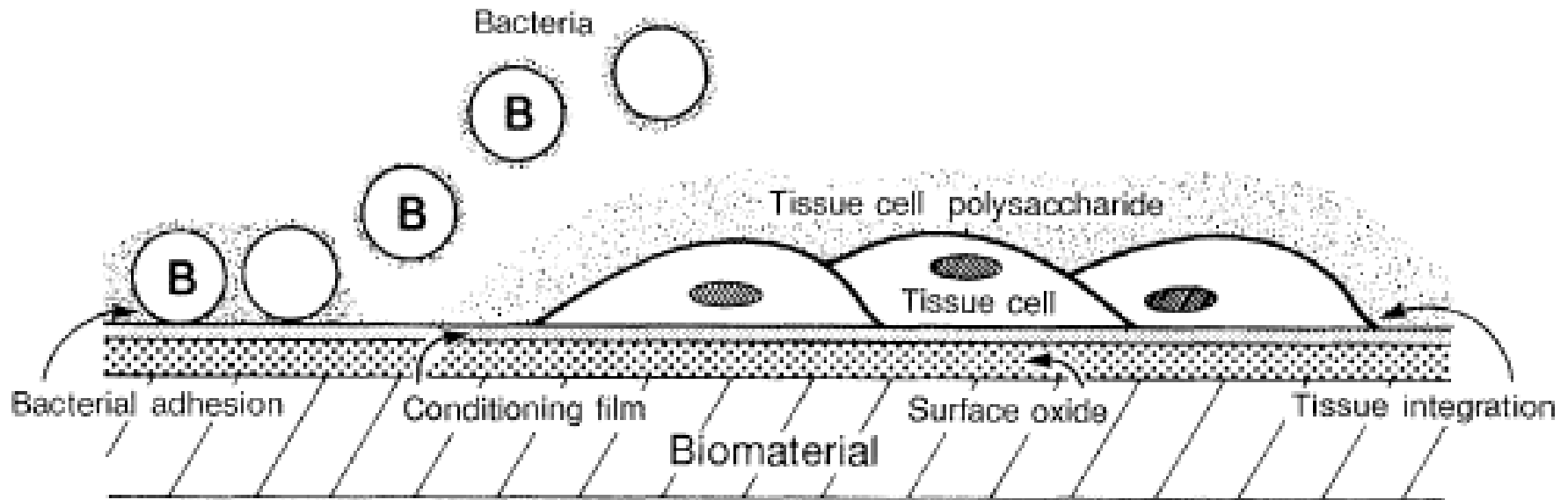
Harmful to the patient

- Processes of coagulation and fibrinolysis
- Formation of clots on the surface of the material
 - These can migrate and clot vessels elsewhere!
- Injury to blood cells, causing low levels of blood-borne cells and clinical problems

Implant-Associated Infection

- A full understanding of host—environment interactions in the pathogenesis of infection also requires an appreciation of **how tissue cells respond to biomaterials**.
- **Biomaterials cause significant perturbations** both locally and systemically in recipient milieus.
- A stage is set at the site of implantation on which the players (bacteria, host cells, and organic moieties) **interact and compete** and for which the host is a reactive audience.
- **Infection around biomaterials and damaged tissues** is caused by bacterial adhesion to those surfaces.
- Tissue integration, adhesion (chemical bonding) of tissue cells to a biomaterial has not been programmed by natural evolution.
- The processes of tissue integration and bacteria adhesion **are biochemically parallel, may be competitive, and may be mutually exclusive**.

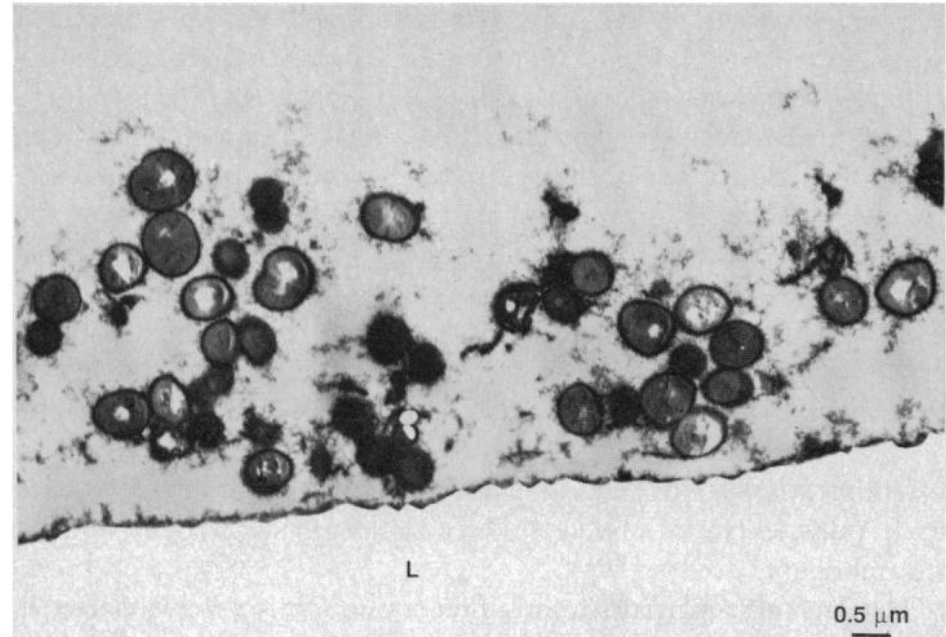
The race for the surface



At the instant of insertion, a biomaterial represents a ready surface for colonization. **It is the nature of biomaterial surfaces that their outer atomic layers must interact instantly with the juxtaposed (并列) biologic environment.** Macromolecules, bacteria, and tissue cells **compete for surface domains at the reactive interface.** The destiny of an implant will be determined by the conditioning macromolecules and cells that dominate its surface. **If the race is won by tissue and a stable integrated relation is achieved, then the surface is less available for bacterial colonization.**

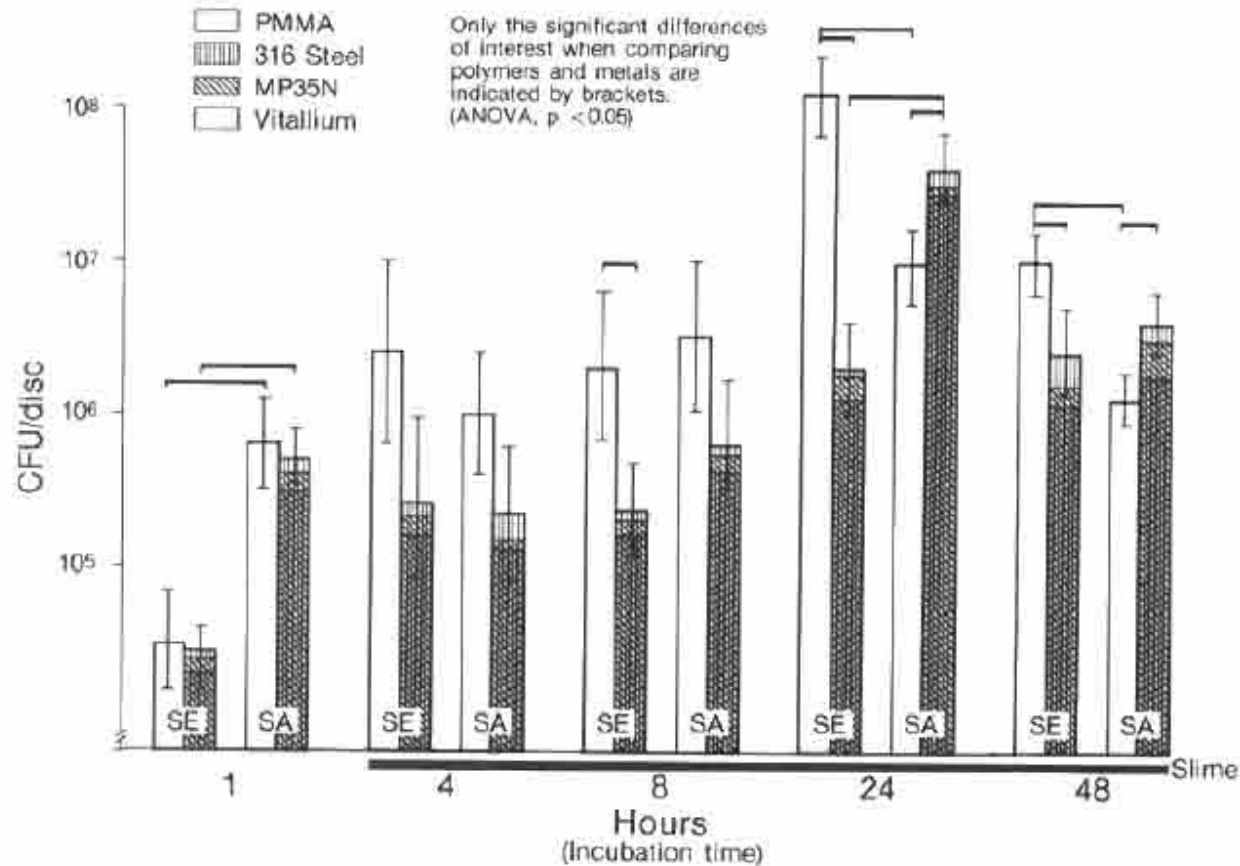
IMPLANT INFECTION (FOREIGN BODY EFFECT)

- (1) a biomaterial or damaged tissue substratum,
- (2) adhesive bacterial colonization of the substratum,
- (3) resistance to host defense mechanisms and antibiotic therapy,
- (4) characteristic bacteria such as *S. epidermidis* and *Pseudomonas aeruginosa*,
- (5) specificity of phenomena (material, organism, host location),
- (6) the transformation of nonpathogens or opportunistic pathogens into virulent organisms by the presence of a biomaterial substratum,
- (7) polymicrobiality,
- (8) persistence of infection until removal of the substratum,
- (9) the absence of tissue integration at the biomaterials-tissue interface,
- (10) the presence of tissue cell damage or necrosis.



Ruthenium-red-stained material encompasses bacteria within clusters, and flocculent material appears to provide continuity with lens surface.

Bacteria Pathogens (clinical and laboratory observations)

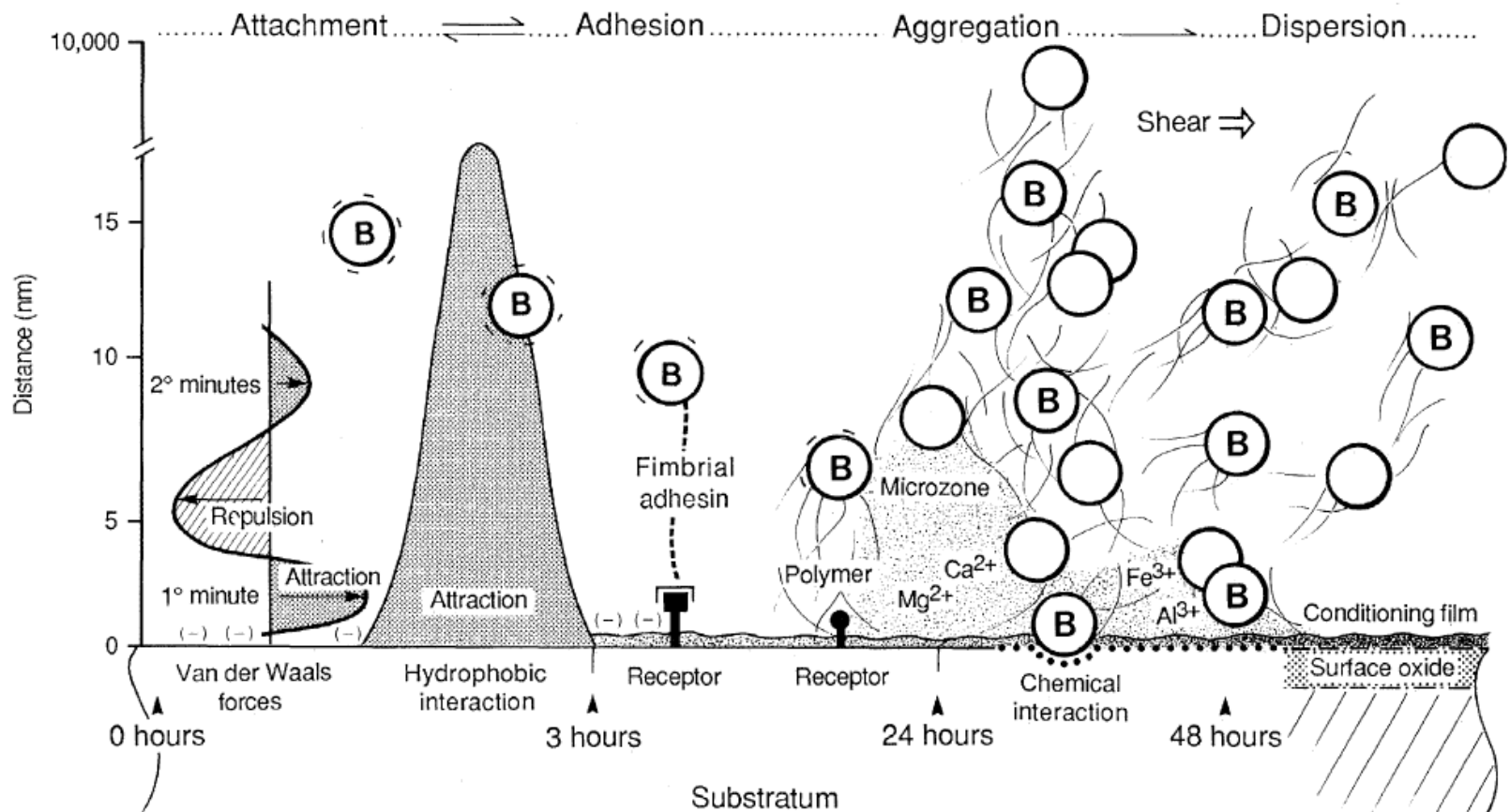


In vitro comparison of colonization of *S. epidermidis* (SE-360) and *S. aureus* (SA-WF1) strains of different biomaterials [i.e., PMMA, multiphase cobalt chromium nickel alloy (MP35N), 316-stainless steel, and vitallium]. At 1 hr, *S. aureus* reaches a temporary maximum. Slime was visible after 4 hr. At 24 hr, there was significantly greater colonization of polymers over metals by *S. epidermidis* and of metals, over polymers by *S. aureus*. Observed maximum colonization was at 24 hours.

Molecular mechanisms of cell-to-surface adhesion

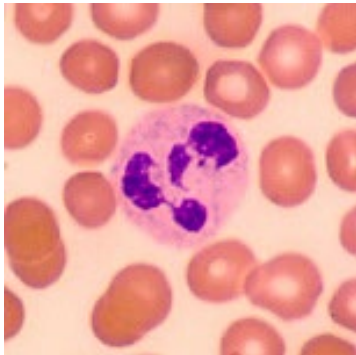
Molecular mechanisms in cell-to-surface adhesion are suggested by **generalizations from colloid chemistry, microbiology, biochemistry, and physics.**

Bacterial particles less than 1 μm in size may be thought initially to behave as in colloid systems.

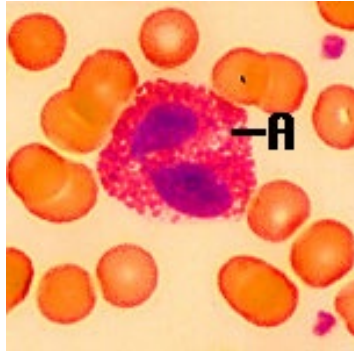


Intravascular Cells (recall)

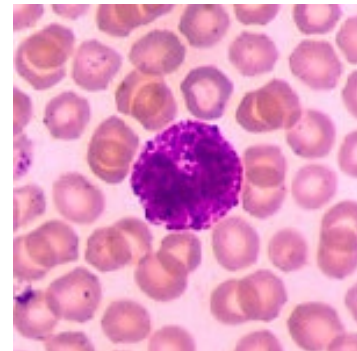
- white blood cells



neutrophil

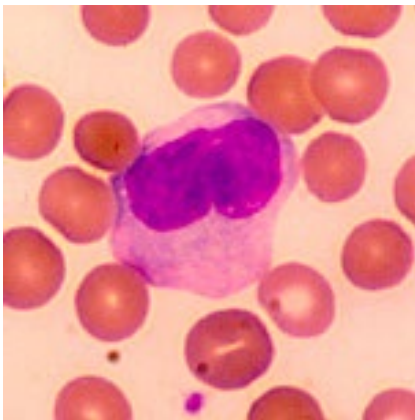


eosinophil

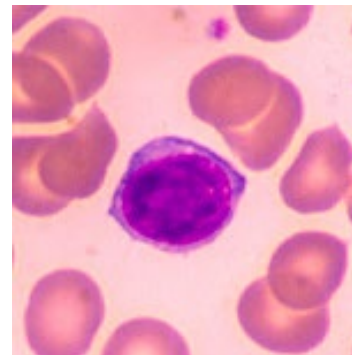


basophil

— Granulocytes



monocyte

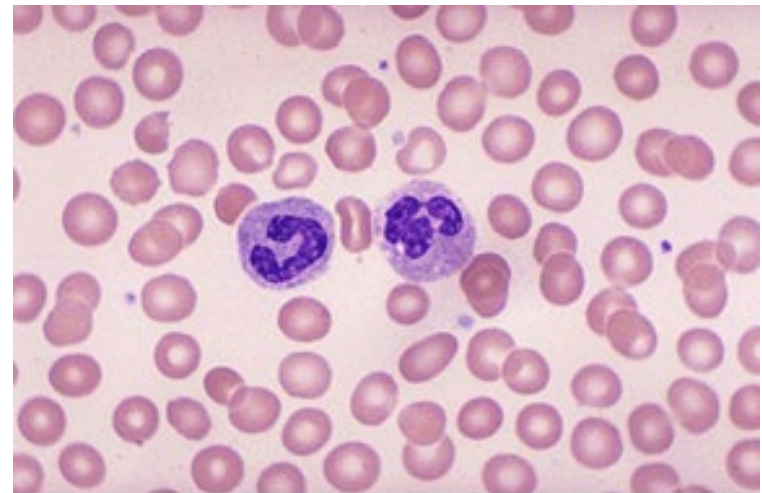


lymphocyte

(recall)

Wright and Giemsa stains are Romanowsky stains used to stain peripheral blood and bone marrow smears. The most important components of these stains are **oxidized methylene blue, azure B and eosin Y dyes**. The eosin Y dye stains the cytoplasm of cells an orange to pink color. The methylene blue and azure B dyes stain the nucleus varying shades of blue to purple. Romanowsky stains are used to perform differential white blood cell counts and to study red blood cell morphology. Abnormal granulocyte, lymphocyte or monocyte cell counts may be used to facilitate the diagnosis of diseases such as leukemia or bacterial infections.

White blood cells stained with Wright's stain:

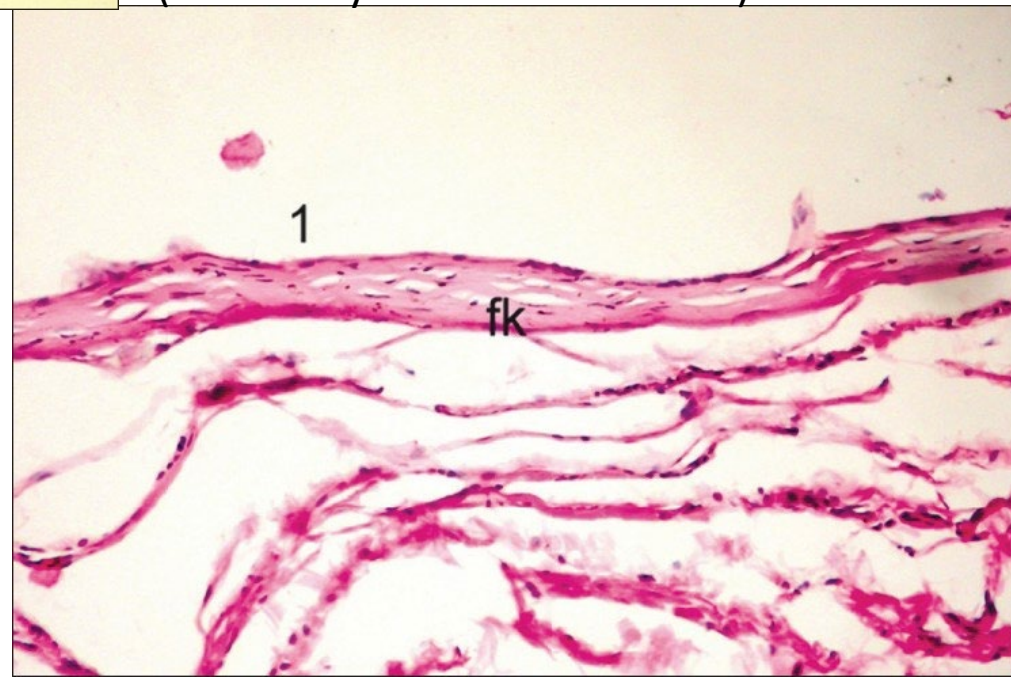


Neutrophils Stained with Wright Stain



Targis: 15th day, moderate inflammatory cell infiltrations (ICI) were observed, where lymphocytes, plasmocytes, and polymorphonuclear leukocytes were present in the connective tissues at the open end of the tubes (1). The presence of fibroblasts, macrophages, and newly formed vessels, as well as foreign body giant cells with engulfed particles of material into their cytoplasm (arrow) were also detected along the short margin (1) of the tube. (2) long margin of the tube (Hematoxylin and eosin: × 10)

Figure 7: Targis: 90th day, a thick fibrous capsule (fk) with decreased number of inflammatory cells along the short margin of the tube (1) (Hematoxylin and eosin: × 10)



Hematoxylin and eosin stain (H&E stain or HE stain) is one of the principal stains in histology. It is the most widely used stain in medical diagnosis and is often the gold standard; for example when a pathologist looks at a biopsy of a suspected cancer, the histological section is likely to be stained with H&E and termed "H&E section", "H+E section", or "HE section".

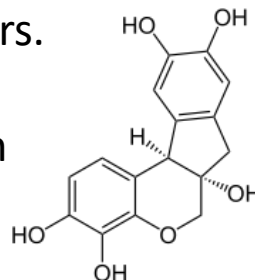
Hematoxylin is a **dark blue or violet stain** that is basic/positive. **It binds to basophilic substances (such DNA/RNA** - which are negatively charged).

DNA/RNA in the nucleus, and RNA in ribosomes in the rough endoplasmic reticulum are both acidic because the nucleic acid building blocks that come off the phosphate backbone are negatively charged. These form salts with basic dyes containing positive charges. Therefore, dyes like hematoxylin will bind to them and stain them violet.

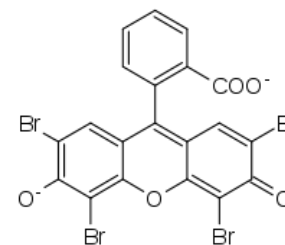
Eosin is **a red or pink stain** that is Acidic / Negative. It binds to acidophilic substances (such as proteins - which are basic and positively charged).

Most proteins in the cytoplasm are basic because they are positively charged due to the arginine and lysine amino acid residues. These form salts with acid dyes containing negative charges, like eosin. Therefore, eosin binds to these amino acids/proteins and stains them pink. This includes cytoplasmic filaments in muscle cells, intracellular membranes, and extracellular fibers.

Hematoxylin



苏木素；苏木精；苏木色素；苏紫



Eosin Y

曙红